An Introduction to Errors and Error Analysis

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Introduction

This resource is created to scaffold your learning of errors and uncertainty in data analysis. Whilst these skills are taught here away from the lab, it is usually in the physical and analytical laboratory setting where you will use these skills.

The resource is written in Markdown and has embedded questions which provide feedback. It is strongly encouraged that you start at the beginning and work through on your first use to check your learning. However you should be able to refer to sections as required when you are more familiar with the content.

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Resource Credits

This resource was created in conjunction with MAST at the University of Bath. Thanks go to Dr Tamsin Smith and in particular Ed Southwood for considerable assistance in developing the code for the regenerative questions.

This workbook works in conjunction with Chapters 2 and 3 of "An Introduction to Contextual Maths in Chemistry" by Fiona Dickinson and Andrew McKinley, which is available in paper and digital form from the University of Bath library.

Error reporting

Whilst care is taken in production of this manuscript, errors may be present. If you find an error, please report using the form below. Thank you for your help in improving this resource.

Resource history

- 16th June 2026 - First commit of index and chapter 1.

1 How many significant figures are there?

1.1 Before we begin... Precision and Accuracy.

In talking about significant figures, standard erro and propagation of error, we need to be clear about what we mean by precision and accuracy.

Data has a spread, in taking physical measurements this is not something we can avoid. How well we can record a value is frequently limited by the equipment we are using. We will talk more about this spread of data in later chapters of this resource.

> Tip 1
>
> - Accuracy is how close a measurement is to the true value.
> - Precision is how close a set of measurements are to each other.

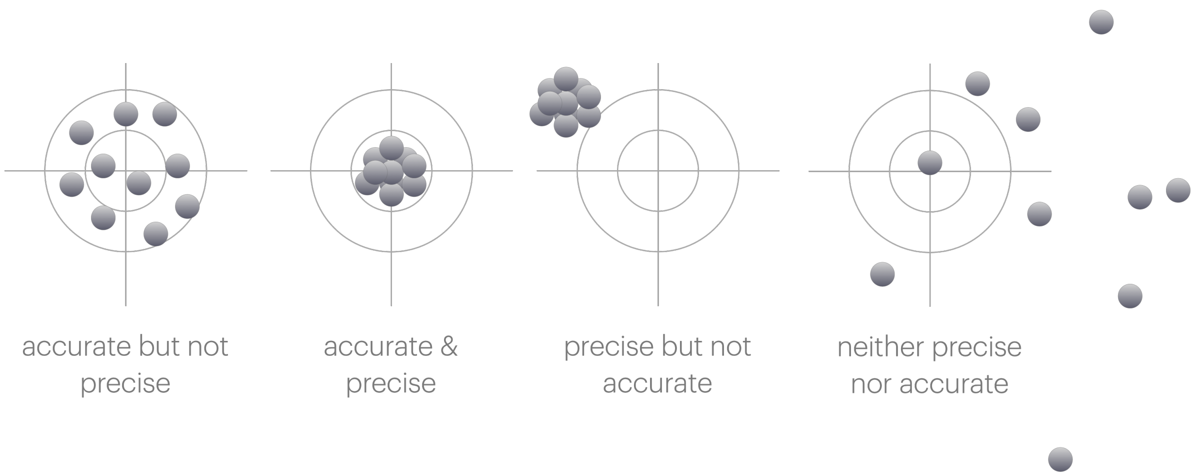

Figure 1.1: The difference between accuracy and precision.

When we are talking about significant figures, we are talking about precision. The more significant figures we have, the more precise our measurement is.

1.2 Significant Figures

Often at school you haven't really thought about significant figures, but in science they are very important. The number of significant figures in a measurement is a measure of how precise that measurement is. The more significant figures, the more precise the measurement.

Therefore it is important that we don't ignore numbers which are significant, or report values to an undue number of significant figures.

Frequently the only way you have thought about significant figures in school is to report things like (2dp) after a value - usually because you have trunkated the value and you want to be clear you are reporting a trunkated value. However this is not appropriate for post school life. Nobody has ever written 2dp or 3sf after a value in a research paper.

> **!** Don't do this!
>
> Never write 2dp or 3sf after a value. It is not appropriate in science and will make you look like you don't know what you are doing.
> Instead just write your answer to the correct number of significant figures.

Therefore we have to have have a way of understanding the significant figures we have, how to report them in an appropriate way, and then we will see in Chapter 3 how to use them in calculations.

> **i** Why we care about significant figures.
>
> Significant figures are a measure of our confidence in a value.
> When we have many significant figures, we are more confident in the value. When we have few significant figures, we are less confident in the value.
> Usually if we have any uncertaintiy in a value it is in the last significant figure we report.

1.3 How many significant figures are there?

Like many things in science, there are rules for how to determine the number of significant figures in a value. The rules are as follows:

> Tip 2
>
> - Any non-zero digit is significant
> - Any 0 following a non-zero digit is significant
> - Any 0 to the left of the first non-zero digit is not significant

By saying that any 0 after another digit we avoid ambiguity. This may be different to what you have been taught in school.

Below are examples of numbers and how many significant figures they have.

> How many significant figures are there?
>
> | | |
> |---|---|
> | 123 | 3 sf |
> | 123.4 | 4 sf |
> | 0.123 | 3 sf |
> | 123400 | 6 sf |
> | 123.4×10^3 | 4 sf |

We should use standard form (or unit prefixes where appropriate) to represent values to an appropriate precision.

> **!** Don't do this!
>
> Freqently in exams I will see answers such as:
> $\Delta H = 123854.3628$ J mol^{-1}
> ≈ 124000 J mol^{-1} (3sf)
> This latter value is actually still written to 6 significant figures.
> We do not need the $\approx$ sign as if are answer should be to three sig figs then the answer is correct to the correct number of significant figures.
> We should write the answer with a unit prefix (or in standard form) to avoid ambiguity.
> $\Delta H = 124$ kJ mol^{-1}

2 Exercise on counting signficant figures

2.1 Reading values

When we record values in the laboratory we are limited by the precision of the instrumentation we are using. We ususally make decisions about equipment unconciously, for example you don't go seraching for a 4 decimal place balance to weigh out most reagents in the synthetic lab, but you frequently do this in the physical lab.

The balances are a good place to remind us that signficant zeros matter.

There is a big diference between 0.4 g and 0.4000 g in terms of our precision.

One thing that may be different to school is accepting that you cannot read a scale more accurately than the scale allows.

Frequently in school you are told that if it is half way between the graduations on a burrette then you use an increased level of precision than the instrument allows. This is not appropriate in science.

Tip 7

You cannot read a scale more accurately than the scale allows.

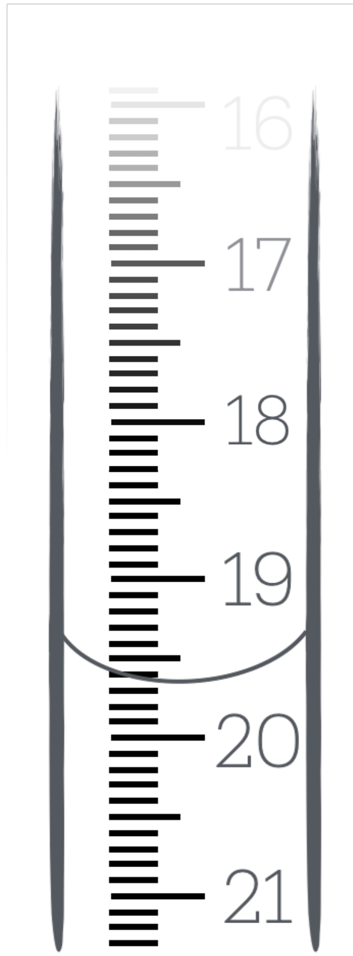

Figure 2.1: The limit of precision on a burette means we can only read as accurately as the scale allows.

In the case of the burette above, we can only read to 1 decimal place. If the meniscus is between 19.6 and 19.7 mL, we **cannot** report the value as 19.65 mL (as you may have been taught in school). This is because we are adding a precision we do not have. How do we know it is 19.65 mL and not 19.66 mL or 19.64 mL? We don't. Therefore we stick with the precision of the instrument.

In this case you either have to choose 'is it closer to 19.6 mL or 19.7 mL' and report the value as either 19.6 mL or 19.7 mL or to avoid systematic bias we *round to even.*

2.2 Round to even

If we think about rounding we need to consider the possibility of systematic bias. If we always round up, then we will always be reporting a value that is higher than the true value. If we always round down, then we will always be reporting a value that is lower than the true value.

Consider the following values which we need to report to 1 decimal place:

Rounding values?

19.10	19.1	no rounding needed
19.11	19.1	round down
19.12	19.1	round down
19.13	19.1	round down
19.14	19.1	round down
19.15	19.?	???
19.16	19.2	round up
19.17	19.2	round up
19.18	19.2	round up
19.19	19.2	round up
19.20	19.2	no rounding needed

If we always round the value ending in a 5 up (as you maybe always have before) then we have more values rounding up than we have rounding ndown and we introduce a slight systematic bias increasing our value over all.

Therefore if a value ends exactly in a 5 and needs to be rounded we instead 'round to even', which should mean that half of the time it will round up and half of the time it rounds down.

Tip 4

If a value ends in a 5 exactly and needs rounding we should round to the nearest even number.

Examples of rounding to even.

If each of the following should be reported to 3 significant figures

47.58	rounds up to	47.6	
8.325	rounds down to	8.32	rounded to even
3.621	rounds down to	3.62	
64.55	rounds up to	64.6	rounded to even
10.050003	rounds up to	10.5	

2.3 Self assessment exercise: write the following values to the listed sf. {sec-SAQroundtoeven}

3 The use of the correct number of significant figures in calculations.

In Chapter 1 we have learnt about significant figures and why they are important in understanding the confidence we have in a value.

Now we need to learn the rules to how many significant figures we should use after doing calculations.

There are different rules for how we handle significant figures depending upon the type of calculation we have done. Freqently we may have more than one step in a calculation, and therefore we need to be concious of the appropriate significant figures in each step.

3.1 Significant figures in addition and subtraction

When adding or subtracting values, the number of decimal places in the result should be the same as the value with the least number of decimal places in your calculation.

We cannot add precision, I can't measure something with a meter rule with 1 cm increments, a 15 cm rule with 1 mm increments and a micrometer with 1 m increments and then add them together and expect to have a result with 1 m precision. The result will be limited by the least precise measurement.

Tip 7: Significant figures in addition and subtraction.

When adding or subtracting values, the number of decimal places in the result should be the same as the value with the least number of decimal places in your calculation.

Examples of significant figures in addition and subtraction.

- Convert 20 ºC to Kelvin.

We know our measured temperature to 0 decimal places, therefore our answer should be to 0 decimal places.
T = 20 ºC + 273.15 K = 293 K

- An ideal gas was heated with 1.75 kJ of energy an allowed to do 250 J of work.

What is the change in internal energy of the gas?

In this case we have different units, so if we convert 250 J to kJ we have 0.250 kJ. The least number of decimal places is 2, therefore our answer should be to 2 decimal places.
$\Delta U = 1.75 \text{ kJ} - 0.250 \text{ kJ} = 1.50 \text{ kJ}$
This unit conversion kJ to J (or vice versa) is common in thermodynamics calculations where ΔH is usually reported in kJ mol^{-1}, but ΔS is usually reported in J K mol^{-1}.
It is easy to forget the significant zero at the end of this number as usually our calculator will not report it, but including it gives us our answer to a higher, more appropriate, precision.

3.2 Exercises with addition subtraction examples.

3.3 Significant figures in multiplication and division

When multiplying or dividing values, the number of significant figures in the result should be the same as the value with the least number of significant figures in your calculation.

First we need to quickly count the number of significant figures in each value. If we have a unit conversion (such as mm to cm) then this is 'exact' and does not change the number of significant figures in our value. Some unit conversions have limited significant figures, others are precise. In either case it is always a good habit to use more significant figures in constants than in our other values.

Tip 7: Significant figures in multiplication and division.

When multiplying or dividing values, the number of significant figures in the result should be the same as the value with the least number of significant figures in your calculation.

Example of significant figures in multiplication and division - the ideal gas law.

- Determine the number of moles of gas in a cylinder with a volume of exactly 2.52 L at a pressure of 50.24 bar and a temperature of 273 K.

Here we need to do a couple of unit conversions to get our values into base SI units.
The volume is already in L, which is a derived unit, but we need to convert it to m^3. 1 L $\equiv$ 1 dm^3 = 1 x 10^{-3} m^3, so we have 2.52 L = 2.52 x 10^{-3} m^3. This is an exact conversion, so the number of significant figures in our value does not change. Recall that the volume is described as "exactly" 2.52 L, so we have 'infinite' significant figures in our volume.
The pressure is in bar, which is not an SI unit, so we need to convert it to Pa. 1 bar = 1

x 10^5 Pa, so we have 50.24 bar = 5.024 x 10^6 Pa. This is also an exact conversion. The pressure has 4 significant figures, so our pressure in bar will also have 4 significant figures.
Using the ideal gas law, we can calculate the number of moles of gas in the cylinder.
$pV = nRT$ so $n = \frac{pV}{RT}$
Putting in our values:

$$n = \frac{5.024 \times 10^6 \text{ Pa} \times 2.52 \times 10^{-3} \text{ m}^3}{8.31446 \text{ J K}^{-1} \text{ mol}^{-1} \times 273 \text{ K}}$$

$n = 5.58$ mol
Temperature only has 3 significant figures, so our answer should be reported to 3 significant figures.

Example of significant figures in multiplication and division - heat capacity.

- Determine the energy put into a system when a 1.000 kg block of copper is heated from 20.16 ºC to 60.52 ºC. The specific heat capacity of copper is 385 J kg^{-1} K^{-1}.

We can calculate the energy put into the system using the equation:
$q = mc\Delta T$
First we need to determine the change in temperature, ΔT.
$\Delta T = T_{\text{final}} - T_{\text{initial}}$
$\Delta T = 60.52C - 20.16C = 40.36C$
We've followed the rules for addition and subtraction, so our answer has 2 decimal places, which is the same as the least number of decimal places in our calculation.
Now we can calculate the energy put into the system, q.
$q = 1.000 \text{ kg} \times 385 \text{ J kg}^{-1} \text{ K}^{-1} \times 40.36 \text{ K}$
The mass has 4 significant figures, the specific heat capacity has 3 significant figures, and the temperature change has 4 significant figures. The least number of significant figures in our calculation is 3, so our answer should be reported to 3 significant figures.
$q = 15.5$ kJ
Note: I've changed my units to kJ to make reporting to 3 significant figures easier. If I had left my answer in J, I would have reported it as 15.5 × 10^3 J, which is not as clear (and is more to type).

3.4 Significant figures in logs and lns

Significant figures in logs and lns are a little different to the rules for addition, subtraction, multiplication and division because we have to think about the 'value' when accounting for how many significant figures we have in our answer.

Tip 7: Significant figures in logs and lns.

When taking logs we usually gain 1 significant figure, however if the 'logged value' is between 1 and the base then we do not gain a significant figure.

Practically what does this mean? and why the do we sometimes gain a significant figure? The following example using $\log_{10}$ will help to explain this.

Example of significant figures in logs and lns.

Determine $\log(2.234)$, $\log(22.34)$ and $\log(223.4)$ to the correct number of significant figures.
Our first example is $\log(2.234)$. The value of $\log(2.234)$ is 0.3490831688....
The value we are taking the log of has 4 significant figures, so our answer should have 4 significant figures. Therefore we report our answer as 0.3491.
Now, if I take the log of 22.34 I can also write this as $\log(2.234 \times 10^1)$.
Using our 'rules of logs', we can write this as $\log(2.234) + \log(10^1) = 0.3491 + 1 = 1.3491$.
Since our value of $\log(10^1)$ is an exact value, it can be considered to have infinite significant figures, and we can add these values according to the rules in Section 3.1.
Finally, if I take the log of 223.4 I can also write this as $\log(2.234 \times 10^2)$. The value we are taking the log of has 4 significant figures, so our answer should have 4 significant figures. Therefore we report our answer as 2.3491.
You can see why we gain a significant figure when taking the log of a value greater than 10, but not when taking the log of a value less than 10. The 'value' of the number we are taking the log of is important in determining how many significant figures we have in our answer.
The same principle works if we have very small numbers, for example $\log(0.002234)$ which we can write as $\log(2.234 \times 10^{-3}) = \log(2.234) + \log(10^{-3}) = 0.3491 - 3 = -2.6509$.

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